

AI-Based Smart Traffic Accident **Risk** **Prediction & Prevention System**

A Next-Generation AI Platform for Predictive Traffic Safety, Accident Prevention, and Intelligent Transportation Management

Team Leader: SUBASRI M

Team Members: PRIYADHARSHINI S, SRIKANTH P

Abstract & Motivation

⚠️ The Road Safety Challenge

Road traffic accidents are a major safety crisis driven by excessive speeds, congestion, deteriorating road infrastructure, adverse weather, and unpredictable driver behavior.

Traditional traffic management is strictly **reactive**—responding to crashes after they occur rather than mitigating root causes beforehand.

🧠 The AI-Powered Paradigm Shift

Our proposed system integrates Artificial Intelligence and Machine Learning to process multi-source data streams in real-time, computing precise **Accident Risk Scores**.

Shifting traffic safety from “reaction after accident” to “prediction and prevention before accident.”

Problem Statement



Reactive Limitations

Current frameworks only handle accident detection and emergency dispatch post-incident, offering zero preventive safeguards.



Fragmented Data Silos

Weather metrics, traffic density, historical accident records, and CCTV feeds are stored and monitored across isolated systems.



Lack of Early Warnings

Drivers and municipal authorities lack real-time predictive hazard notifications for dangerous road segments during high-risk windows.



Manual Monitoring Bottlenecks

Human-dependent traffic monitoring fails to scale across expanding urban road networks and complex highways.



Blind Hazard Zones

Accident-prone spots remain unidentified until critical thresholds or fatal clusters accumulate over prolonged periods.



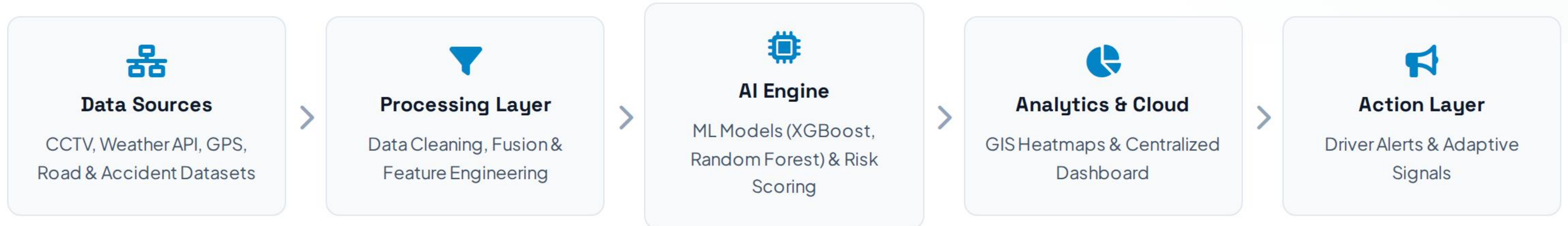
The Solution Need

An intelligent, automated AI-based forecasting engine capable of continuously identifying and mitigating risk before accidents occur.

Existing vs. Proposed System Comparison

Evaluation Criteria	Existing System	Proposed AI System
Core Approach	Reactive (Post-incident response)	Predictive (Proactive prevention)
Data Integration	Isolated sources & manual logs	Multi-source real-time data fusion
Risk Forecasting	Limited / Historical reporting only	Advanced AI/ML predictive analytics
Driver Alerts	Static road signs & delayed warnings	Real-time dynamic hazard notifications
Traffic Control	Fixed-timer or manual intervention	Intelligent, adaptive signal support
Visualization	Basic CCTV monitoring	GIS Heatmaps & Authority Dashboards

System Architecture



Methodology Workflow

STEP 01

Data Ingestion

Collect streaming CCTV footage, weather updates, vehicle speeds, and historical crash data.

STEP 02

Feature Extraction

Isolate key metrics: visibility, traffic density, road surface condition, and time patterns.

STEP 03

AI Risk Modeling

Train ensemble ML models to calculate real-time accident probability scores.

STEP 04

Risk Classification

Categorize zones into Low, Medium, High, and Critical safety tiers.

 Continuous Feedback Loop:

Real-time incident outcomes continuously retrain and optimize model prediction accuracy.

Key Modules & Risk Tiers

Core Modules

- **Risk Prediction Engine:** Computes segment-specific accident probability.
- **CCTV Vision Module:** Detects congestion and hazardous maneuvers.
- **Weather & Road Module:** Factors rain, low visibility, and surface hazards.
- **Smart Alert & Signal System:** Triggers warnings and adaptive signal control.

Risk Classification Levels

Normal Traffic Conditions	LOW RISK
Moderate Congestion / Slower Flow	MEDIUM RISK
Adverse Weather / Speed Anomalies	HIGH RISK
Immediate Crash Probability Warning	CRITICAL RISK

Technology Stack



AI / Machine Learning

Python, Scikit-Learn, XGBoost, TensorFlow, PyTorch, OpenCV, Pandas, NumPy



Backend & APIs

FastAPI / Flask, RESTful APIs, PostgreSQL, MongoDB, Weather & Map APIs



Frontend & Visualization

React, JavaScript, HTML5, CSS3, GIS Mapping APIs, Interactive Risk Dashboards

Advantages & Limitations

✓ System Advantages

- Shifts safety paradigm from reactive to predictive.
- Provides real-time early hazard warnings to drivers.
- Identifies high-risk accident hotspots proactively.
- Optimizes emergency response and municipal planning.
- Scalable architecture deployable in smart cities.

⚠ Current Limitations

- High dependency on robust, quality dataset inputs.
- Requires reliable high-speed network connectivity.
- Infrastructure setup costs for high-density CCTV/sensors.
- Potential false-positive risk warnings in edge cases.

Conclusion & Future Scope

Conclusion

The AI-Based Smart Traffic Accident Risk Prediction and Prevention System successfully transforms urban transportation safety into a proactive science.

“Predict the Risk. Prevent the Accident. Save Lives.”

Future Enhancements

- Integration with Connected Autonomous Vehicles (CAVs).
- Edge AI deployment for instantaneous on-device processing.
- Drone-based dynamic traffic and accident surveillance.
- Vehicle-to-Infrastructure (V2I) real-time communication.

THANK YOU